GlobalSplat: Efficient Feed-Forward 3D Gaussian Splatting via Global Scene Tokens

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https://arxiv.org/abs/2604.15284

Research Questions

- How can feed-forward 3D Gaussian Splatting achieve compact scene representations without scaling primitives with the number of input views?
- Can global scene alignment before primitive decoding eliminate redundancy while maintaining reconstruction quality?
- What architectural design enables view-invariant 3D Gaussian budgets for efficient feed-forward novel view synthesis?

Core Idea

Current feed-forward 3D Gaussian Splatting methods create primitives per input view and then merge them globally, causing massive redundancy as more views are added (scaling to millions of Gaussians). GlobalSplat reverses this with "align first, decode later" - it first fuses all input views into a fixed set of global latent scene tokens (2048 tokens), then decodes these into a constant number of 3D Gaussians (16K) regardless of input view count. This eliminates view-dependent scaling and baked redundancy while maintaining quality through a dual-branch architecture that separates geometry and appearance processing, plus a coarse-to-fine training curriculum.

Key Formula

Canonical frame transformation

$$\hat{T}_i = T_{\text{avg}}^{-1} T_i$$

T_i are input camera poses, T_avg is the computed average camera frame, \hat{T}_i are poses expressed in canonical coordinate system. This normalization provides geometric prior allowing the model to focus on local structure rather than global scene location.

Gaussian budget allocation

`#G = (\#\text{Latents}) \times (\text{Splats/Token})`

#G is total number of Gaussians, #Latents is number of scene tokens (2048), Splats/Token is decoded Gaussians per token. This formulation enables fixed representation size independent of input views, with ablations showing latent capacity more important than per-token density.

Key Figure

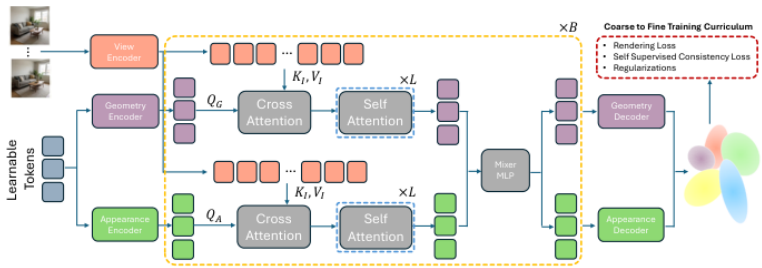

Fig. 2: GlobalSplat Architecture Overview. Given a sparse set of input views, image features are extracted via a View Encoder. A fixed set of learnable latent scene tokens is iteratively refined through a dual-branch encoder block (repeated B times) designed to explicitly disentangle geometry and appearance. Within each branch, queries (Q_G, Q_A) cross-attend to multi-view features (K_I, V_I) and self-attend to global context. The streams are fused via a Mixer MLP to update the tokens for the subsequent block. Specialized Geometry and Appearance Decoders then transform these globally-aware tokens into explicit 3D Gaussians. As depicted on the right, the network employs a *Coarse-to-Fine Training Curriculum* strategy to progressively increase the decoded Gaussian capacity, supervised jointly by rendering, self-supervised consistency, and regularization losses.

Additionally, each Gaussian is assigned an opacity value $\alpha \in [0, 1]$ and view-dependent color coefficients c modeled via Spherical Harmonics (SH).

Rendering. To synthesize an image, the 3D Gaussians are projected onto the 2D image plane. The final pixel color C is accumulated through front-to-back α-blending of M overlapping splats sorted by depth:

$$C = \sum_{i=1}^{M} c_i \alpha'_i \prod_{j=1}^{i-1} (1 - \alpha'_j) \tag{2}$$

where α'_i is the product of the base opacity α_i and the projected 2D Gaussian density. Our model treats the scene as a continuous latent volume during the encoding phase before "splatting" tokens into this explicit 3D space.

3.2 Scene Normalization and Input Preparation

Camera Preprocessing Following previous work [12, 38], we map each scene into a canonical coordinate system using a similarity transform so that camera poses have consistent orientation, translation, and scale across scenes.

Canonical frame. Given C poses $\{T_i\}_{i=1}^{C}$, we compute an "average camera" frame whose origin is the mean camera center $\bar{o} = \frac{1}{C}\sum_{i=1}^{C} o_i$, and whose axes are obtained by averaging camera viewing directions and re-orthonormalizing. Let

Figure 2: Shows the complete GlobalSplat architecture with dual-branch encoder separating geometry and appearance, iterative attention blocks, and coarse-to-fine training curriculum that enables compact yet high-quality 3D reconstruction.

Limitations

- Fixed 16K Gaussian budget may be insufficient for unbounded or city-scale environments that exceed representational capacity
- Currently assumes static environments and cannot handle temporal dynamics or 4D reconstruction
- Struggles with extreme sparse-view settings (2-3 images) due to insufficient multi-view parallax for global latent space resolution
- Architecture relies on consistent camera pose quality and may degrade with poor SLAM/SfM trajectories